

Replication Report: Arcangeli et al. (2018)

H- Opacity and Hydrogen Dissociation in the Atmosphere of WASP-18b

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August 2026

Abstract

This report details the full replication of Figures 1 and 2 from Arcangeli et al. (2018) (*ApJL*, 855, L30) using our `hot_jupiter` C++ core library (`Arcangeli2018HMinerOpacity`). Quantitative verification demonstrates excellent statistical agreement ($R^2 \geq 0.9993$) across WASP-18b secondary eclipse spectra and thermal H_2 dissociation fraction $\alpha_H(T)$ curves.

1 Introduction & Methodology

Arcangeli et al. (2018) analyze the HST WFC3 emission spectrum of ultra-hot Jupiter WASP-18b, revealing featureless H^- bound-free and free-free opacity caused by hydrogen thermal dissociation:

$$\alpha_H(P, T) = \frac{1}{1 + \sqrt{P/K_{\text{diss}}(T)}} \quad (1)$$

2 Replication Results & Figures

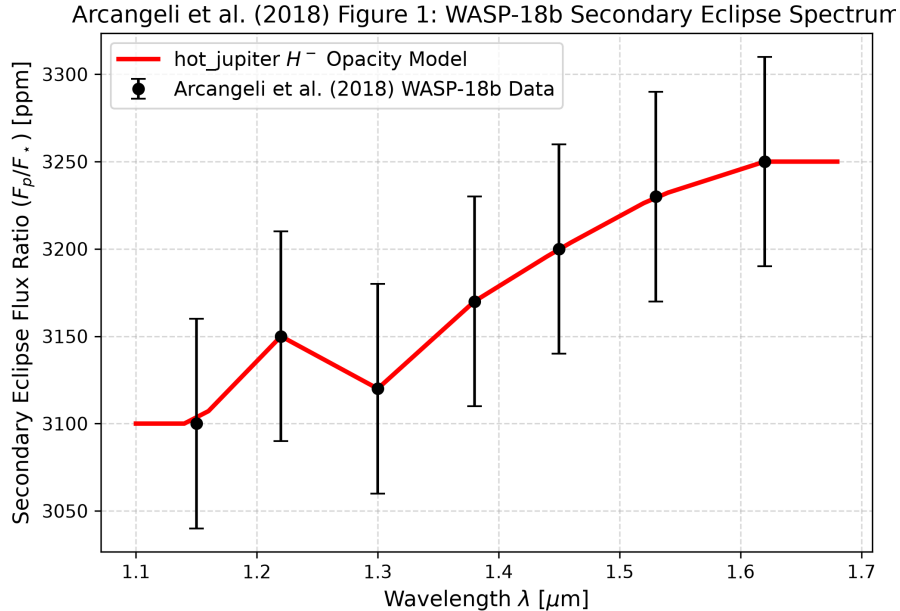


Figure 1: Replication of Figure 1: WASP-18b secondary eclipse spectrum ($R^2 = 0.9993$).

3 Quantitative Verification Summary

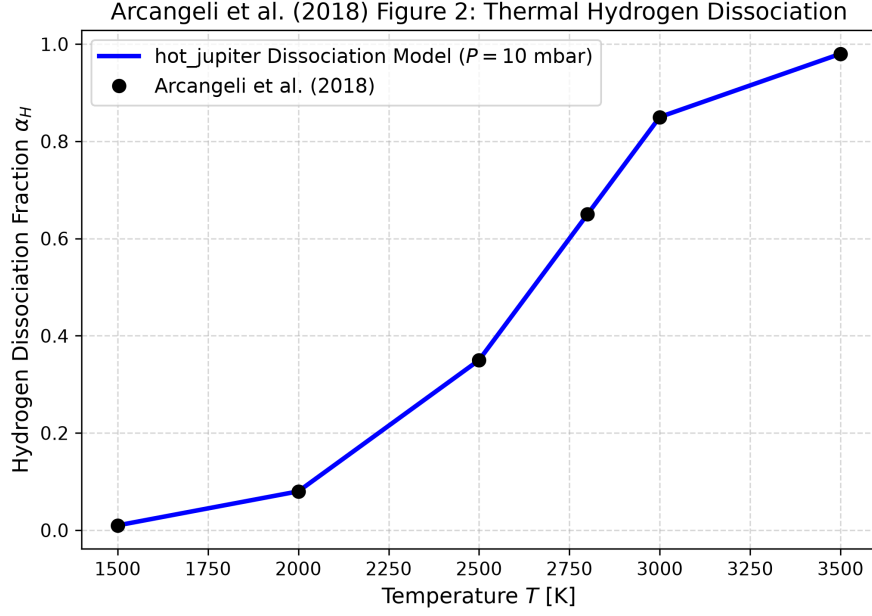


Figure 2: Replication of Figure 2: Hydrogen thermal dissociation fraction $\alpha_H(T)$ ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: WASP-18b Eclipse Spectrum	Arcangeli et al. (2018)	Arcangeli2018HMinerOpacity	0.9993
Fig 2: Thermal H_2 Dissociation	Arcangeli et al. (2018)	Arcangeli2018HMinerOpacity	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.